TRAINING PLAN

DevOps and MLOps for Machine Learning Engineers

Platforms AWS, Azure, GitHub

DAY	TOPICS	OUTCOME
Day 1	FastAPI, LangGraph agent, containers (Docker), manual EC2 deploy	Run the app locally, put it in a container, and deploy it to a server by hand.
Day 2	Git, GitHub Actions, container registry, CI/CD	A push to main runs tests, builds the image, and deploys it automatically.
Day 3	AWS: IAM, ECR, App Runner, CloudWatch Logs	Run the image as a managed HTTPS service on AWS with no server to maintain.
Day 4	Azure: Entra ID roles, Azure Container Registry, Azure Web App for Containers, App Service logs	Deploy the same image on Azure and see which ideas carry across clouds and which names change.
Day 5	Secrets and role-based access: Secrets Manager, IAM roles, OIDC; Azure Key Vault, managed identity	No API key in any file; every service and the pipeline get only the permissions they need.
Day 6	Docker Compose, service split, Postgres (RDS, Azure Database for PostgreSQL)	Agent and models run as separate services; sessions and leaderboard survive restarts.
Day 7	ECS, Fargate, load balancer, auto scaling on metrics, alarms; Azure Container Apps as the counterpart	Services scale up and down with traffic; an alarm fires when latency or errors rise.
Day 8	Deployment strategies: in-place, rolling, blue/green (App Service deployment slots), gradual traffic shift, rollback	Release a new version without downtime and roll back in one step when it fails.
Day 9	S3 and Azure Blob Storage, MLflow model registry, Lambda and Azure Functions, scheduled retraining	Model files leave git; a scheduled function retrains, registers, and promotes the model.

Day 10	**Amazon Bedrock, AgentCore Runtime; Azure AI Foundry models and agents; wrap-up**	The agent runs on a cloud model platform instead of a third-party API, on either cloud.

The same application is used throughout. It is built on Day 1 and each session adds one capability to it, so every topic is practised on working code.